GENOMICS CLINICAL ACTION REPORT & PREVENTIVE SURVEILLANCE SUMMARY

Patient: Daniel Ehrle (Daniel_Ehrle) · **Date:** October 12, 2026

Primary Audience: Primary Care Physician (PCP) (Annual Physical, Preventive Lab Screening & Prescribing) · **Co-Routing:** Cardiology, Oncology, Hematology, Neurology & Clinical Pharmacy

Scope: 1407 Curated Variants (1002 Genes) + 41 Pharmacogenes

SECTION 1 · PRIORITY GENOMIC FINDINGS, CARRIER STATUS & CLINICAL IMPACT

GENE, SYSTEM & INHERITANCE	VARIANT, RSID & COORDINATE	ZYGOSITY & DEPTH	CLASSIFICATION & SCORES	CLINICAL IMPACT, PRIMARY CARE ACTION & ZYGOSITY CONTEXT
CBLIF Multisystem **Autosomal recessive**	**c.79+1G>A · rs147785187** chr11:59845374	**Heterozygous** DP: 49x	**Pathogenic** CADD 32.0 · AVI 33.9 · Conf 0.95	**CBLIF · Ileal Cobalamin (Vitamin B12) Absorption & One-Carbon Metabolism:** Canonical splice-donor variant (c.79+1G>A) in gastric intrinsic factor (autosomal recessive **Hereditary Intrinsic Factor Deficiency, OMIM #261000**), which mediates ileal cubam-receptor endocytosis of dietary Vitamin B12. **Clinical Impact:** Heterozygous carriers preserve intrinsic factor secretion from the wild-type allele; periodic serum Vitamin B12 (> 350 pg/mL) and methylmalonic acid (MMA) verify cellular cobalamin stores (if subclinical depletion occurs, oral/sublingual B12 1,000 mcg/day restores levels via passive mucosal diffusion). **Heterozygous Recessive Carrier:** Single-allele carrier with an intact wild-type parental allele (~50–100% baseline functional reserve); does not cause primary recessive disease and requires no invasive diagnostic workup.
GJB2 Multisystem **Autosomal recessive**	**p.Met34Thr · c.101T>C · rs35887622** chr13:20189481	**Heterozygous** DP: 57x	**Pathogenic** CADD 20.9 · AVI 23.6 · Conf 0.99	**GJB2 · Cochlear Gap Junction & Auditory Potassium Homeostasis:** Coding variant (p.Met34Thr) in cochlear gap junction protein Connexin-26 (DFNB1A, OMIM #220290), which recycles endolymphatic potassium ions in hair cells. **Clinical Impact:** Heterozygous carriers typically maintain normal hearing thresholds; establish a baseline pure-tone audiogram and exercise caution with ototoxic aminoglycoside antibiotics (gentamicin, tobramycin, amikacin) when alternative antimicrobials exist. **Heterozygous Recessive Carrier:** Single-allele carrier with an intact wild-type parental allele (~50–100% baseline functional reserve); does not cause primary recessive disease and requires no invasive diagnostic workup.
VDR Multisystem **Pathway Modifier**	**c.-1172A>G · rs4516035** chr12:47906043	**Homozygous** DP: 49x	**Likely pathogenic** CADD 5.4 · AVI 2.1 · Conf 0.85	**VDR · Calcitriol Receptor Transcription & Bone Mineral Homeostasis:** 5'-UTR promoter regulatory variant (c.-1172A>G) in the nuclear vitamin D3 (calcitriol) receptor (OMIM #601769), which regulates intestinal calcium transport and osteoblast gene transcription. **Clinical Impact:** Modulates baseline calcitriol receptor expression; maintain serum 25(OH)D at 35–55 ng/mL via routine cholecalciferol (Vitamin D3) supplementation. **Homozygous Pathway Modifier:** Modulates baseline pathway or receptor kinetics without causing high-penetrance monogenic disease (manage via routine annual bloodwork; no invasive workup warranted).
F5 Multisystem **Autosomal dominant**	**p.Arg534Gln · c.1601G>A · rs6025** chr1:169549811	**Heterozygous** DP: 58x	**drug response** CADD 27.9 · AVI 23.6 · Conf 0.99	**F5 · Coagulation Cascade & Activated Protein C (APC) Regulation:** Heterozygous Factor V Leiden (p.Arg534Gln) impairs Activated Protein C (APC) cleavage of Factor Va, conferring a ~3–5x relative risk of venous thromboembolism (VTE). **Clinical Impact:** Routine lifelong anticoagulation is **not** indicated in asymptomatic heterozygotes; apply situational perioperative/immobilization VTE prophylaxis and avoid systemic prothrombotic estrogen formulations. **Heterozygous Functional / Pathway Variant:** Single-allele variant with an intact wild-type copy; evaluate physiological expression via routine primary care labs or office screening (no invasive workup warranted).
PEX6	**p.Val788Met ·**	**Heterozygous**	**Conflicting**	**PEX6 · Peroxisomal Biogenesis & Hepatic Very-Long-

SECTION 2 · PREVENTIVE LABORATORY & CLINICAL SURVEILLANCE SCHEDULE (PCP ANNUAL PHYSICAL & SPECIALIST ROUTING)

Derived strictly from **Daniel Ehrle**'s variant callset for integration into the **Primary Care Physician (PCP) annual preventive physical**. Each row links the triggering **Gene, Variant, and Zygosity** to a specific laboratory panel or clinical evaluation. (Page 2 of 3)

GENE, VARIANT & ZYGOSITY	PHYSIOLOGICAL PATHWAY	LABORATORY TEST / EVALUATION	ORDERING CLINICIAN & CO-CONSULT	FREQUENCY	CLINICAL RATIONALE & THRESHOLDS
CBLIF **c.79+1G>A · rs147785187** chr11:59845374 · DP: 49x **Heterozygous (Autosomal recessive)**	**CBLIF · Ileal Cobalamin (Vitamin B12) Absorption & One-Carbon Metabolism** ClinVar: Pathogenic	**Serum Vitamin B12 (Cobalamin), Methylmalonic Acid (MMA), Homocysteine & CBC with MCV** Venous Blood Draw	**Primary Care Physician (PCP)** Co-Consult: Hematology / GI (only if lab deficiency confirmed)	Baseline at Annual PCP Physical; repeat every 1–2 years	Canonical splice-donor variant (c.79+1G>A) in gastric intrinsic factor (autosomal recessive **Hereditary Intrinsic Factor Deficiency, OMIM #261000**), which mediates ileal cubam-receptor endocytosis of dietary Vitamin B12. **Clinical Impact:** Heterozygous carriers preserve intrinsic factor secretion from the wild-type allele; periodic serum Vitamin B12 (> 350 pg/mL) and methylmalonic acid (MMA) verify cellular cobalamin stores (if subclinical depletion occurs, oral/sublingual B12 1,000 mcg/day restores levels via passive mucosal diffusion).
GJB2 **p.Met34Thr · c.101T>C · rs35887622** chr13:20189481 · DP: 57x **Heterozygous (Autosomal recessive)**	**GJB2 · Cochlear Gap Junction & Auditory Potassium Homeostasis** ClinVar: Pathogenic	**Baseline Pure-Tone Audiometry & Ototoxic Antibiotic Precaution** Office Audiogram	**Primary Care Physician (PCP)** & **Audiology / ENT**	Baseline audiogram at routine preventive visit; repeat if auditory symptoms occur	Coding variant (p.Met34Thr) in cochlear gap junction protein Connexin-26 (DFNB1A, OMIM #220290), which recycles endolymphatic potassium ions in hair cells. **Clinical Impact:** Heterozygous carriers typically maintain normal hearing thresholds; establish a baseline pure-tone audiogram and exercise caution with ototoxic aminoglycoside antibiotics (gentamicin, tobramycin, amikacin) when alternative antimicrobials exist.
VDR **c.-1172A>G · rs4516035** chr12:47906043 · DP: 49x **Homozygous (Pathway Modifier)**	**VDR · Calcitriol Receptor Transcription & Bone Mineral Homeostasis** ClinVar: Likely pathogenic	**Serum 25-Hydroxyvitamin D [25(OH)D], Total Calcium, Albumin & Alkaline Phosphatase** Venous Blood Draw	**Primary Care Physician (PCP)** Co-Consult: Endocrinology (if refractory deficiency)	Annual PCP Preventive Lab Panel	5'-UTR promoter regulatory variant (c.-1172A>G) in the nuclear vitamin D3 (calcitriol) receptor (OMIM #601769), which regulates intestinal calcium transport and osteoblast gene transcription. **Clinical Impact:** Modulates baseline calcitriol receptor expression; maintain serum 25(OH)D at 35–55 ng/mL via routine cholecalciferol (Vitamin D3) supplementation.
F5 **p.Arg534Gln · c.1601G>A · rs6025** chr1:169549811 · DP: 58x **Heterozygous (Autosomal dominant)**	**F5 · Coagulation Cascade & Activated Protein C (APC) Regulation** ClinVar: drug response	**Situational Perioperative VTE Risk Review & Thromboprophylaxis Planning** Perioperative / Immobilization Assessment	**Primary Care Physician (PCP)** Co-Consult: Hematology / Perioperative Medicine	Situational (Prior to major surgery, prolonged immobilization, or hormone therapy)	Heterozygous Factor V Leiden (p.Arg534Gln) impairs Activated Protein C (APC) cleavage of Factor Va, conferring a ~3–5x relative risk of venous thromboembolism (VTE). **Clinical Impact:** Routine lifelong anticoagulation is **not** indicated in asymptomatic heterozygotes; apply situational perioperative/immobilization VTE prophylaxis and avoid systemic prothrombotic estrogen formulations.
PEX6 **p.Val788Met · c.2362G>A · rs267608240** chr6:42966044 · DP: 48x **Heterozygous**	**PEX6 · Peroxisomal Biogenesis & Hepatic Very-Long-Chain Fatty Acid Oxidation**	**Hepatic Function Panel (AST, ALT, Bilirubin) & Fasting Lipid Profile** Venous Blood Draw	**Primary Care Physician (PCP)**	Annual PCP Preventive Blood Panel	Missense variant (p.Val788Met) in Peroxisomal Biogenesis Factor 6 (AAA ATPase matrix protein importer, OMIM #601498). **Clinical Impact:** Single-allele heterozygous carriers maintain normal peroxisomal

SECTION 4 · ACTIONABLE PHARMACOGENOMIC PRESCRIBING GUARDRAILS (CPIC / DPWG MEDICATION & DIPLOTYPE REFERENCE)

PRESCRIBING ACTION TIER	AFFECTED MEDICATIONS BY CAUSAL PHARMACOGENE & DIPLOTYPE	CLINICAL MECHANISM & PRESCRIBING RULE
🚫 AVOID / CONTRAINDICATED 2 Screened Drugs	**F5** [Normal Metabolizer]: **Combined Oral Contraceptives (Estrogen/Progestin), Hormone Replacement Therapy (Estrogen Hrt)**	High risk of toxicity or therapeutic failure; select an alternative medication. *Example (Combined Oral Contraceptives (Estrogen/Progestin)):* Factor V Leiden carrier (rs6025 C>T heterozygous). 3- to 8-fold increased baseline venous thromboembolism (VTE) risk, amplified up to 35-fold with estrogen-containing contracept…
⬇️ REDUCE DOSE (DOSE DOWN) 6 Screened Drugs	**CYP2C9** `(*1/*2)` [Intermediate Metabolizer]: **Celecoxib (Celebrex), Phenytoin (Dilantin), Warfarin (Coumadin)** · **CYP2C19/NR1I2/POR/ABCB1** `(*1/*17)` [Rapid Metabolizer]: **Voriconazole** · **MTHFR/DPYD/RGS5/SELE/EXO1/UGT1A4/OGG1/XPC/KDR/MSH3/TMEM167A/XRCC4/FLT4/HLA-G/VEGFA/SOD2/ABCB1/PON1/NOS3/WRN/POU5F1B** `(*1/*1)` [Normal Metabolizer]: **Fluorouracil** · **CYP3A5** `(*3/*3)` [Poor Metabolizer]: **Ondansetron**	Impaired enzymatic clearance or transporter function increases systemic drug exposure; reduce starting/maintenance dose. *Example (Celecoxib (Celebrex)):* Moderate decrease in celecoxib clearance. Initiate therapy at lowest recommended dose and monitor for GI/cardiovascular adverse effects.
⬆️ INCREASE DOSE / RAPID CLEARANCE 2 Screened Drugs	**CYP2C19** `(*1/*17)` [Rapid Metabolizer]: **Omeprazole / Pantoprazole / Lansoprazole** · **CYP3A5** `(*3/*3)` [Poor Metabolizer]: **Vardenafil**	Rapid/ultrarapid enzymatic clearance may cause subtherapeutic exposure at standard doses. *Example (Omeprazole / Pantoprazole / Lansoprazole):* Increased clearance of proton pump inhibitors. For H. pylori eradication or severe GERD, consider increasing daily dose 50-100%.
⚠️ USE WITH CAUTION 124 Screened Drugs	**NAT2/NR1I2** `(*5/*5)` [Poor Metabolizer]: **Isoniazid** · **CYP3A5** `(*3/*3)` [Poor Metabolizer]: **4-Hydroxycyclophosphamide** · **UGT1A1** `(*1/*28_80)` [Intermediate Metabolizer]: **Belinostat, Irinotecan (Camptosar)** · **ADRB2/ADRB1** `(*1/*1)` [Normal Metabolizer]: **Beta Blocking Agents** · **HDAC1** `(*1/*1)` [Normal Metabolizer]: **Corticosteroids For Systemic Use** · **IL13** `(*1/*1)` [Normal Metabolizer]: **Hepatitis Vaccines**	Moderate gene-drug interaction; monitor clinical response and adverse effects. *Example (Isoniazid):* NAT2 *5/*5 slow acetylator phenotype. Higher risk of isoniazid-induced hepatotoxicity and peripheral neuropathy. Monitor liver function tests (LFTs) and administer pyridoxine